

Epidemics in Queueing

Analyzing the Spread of Infection in the M/M/1 Queue

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Background

Epidemics in
Queueing

Newman

Model

Probability you
get infected

Number of people
you infect

Fluctuations in
that number

Numerical results

References

U.S. coronavirus deaths in five different forecasts

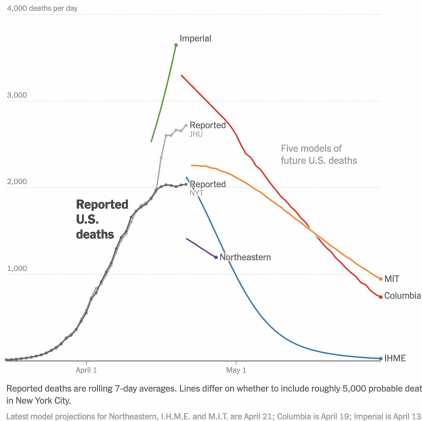


Figure: NYT April 2020 Reporting on Covid Modelling

<https://www.nytimes.com/interactive/2020/04/22/upshot/coronavirus-models.html>

Motivation

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- Current stochastic models assume homogeneity across populations.
- Research allowing for heterogeneity centers around simulations (Ying and O'Clery [9], Namila et al. [6], Agnelli et al. [2], Johansson and Goscè [4], Ronchi and Lovreglio [8], Goscé et al. [3], Abadeer and Gorlatch [1], Rahn et al. [7].)
- In service systems, customers overlap rather than simultaneous interaction across the entire system.
- Current similar work (Kang et al. [5]) does not take the perspective of determining infection risk for individuals, and combining this perspective with a systemic view of individual infection numbers highlights underlying relationships in the model.
- Need for methodologies which allow for generalization of distributions.

Talk Outline

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1 Model

2 Probability you get infected

3 Number of people you infect

4 Fluctuations in that number

5 Numerical results

Model

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Model

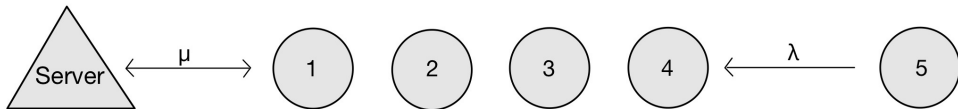
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- Arrival rate λ , Poisson process
- Service rate μ , exponential
- FCFS discipline

Model

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Model

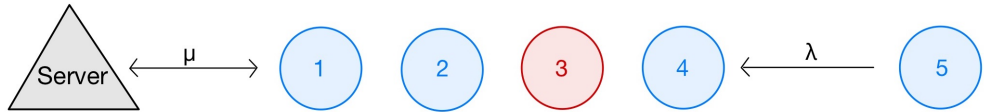
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We suppose an infectious probability p that any customer is already infected upon entering the system, and a susceptible probability $1 - p$ that a customer is healthy.

Model

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Model

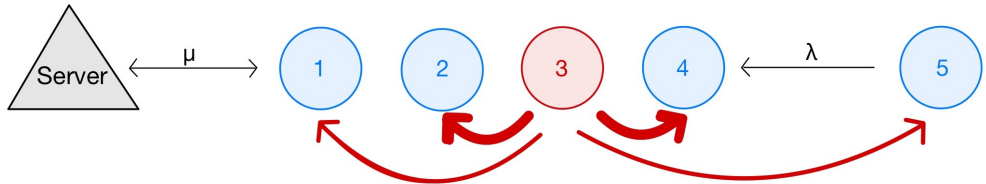
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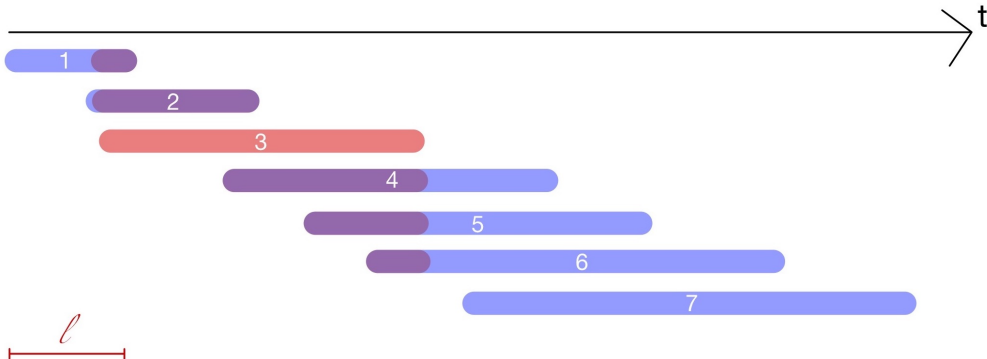
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We suppose that if a susceptible customer is in the system for at least l units of time overlapping with an infectious customer, then the susceptible customer becomes infected.

Methodology

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For an arrival at time t , we can partition all the customers into those who arrive before time t and those who arrive after.

- **Case 1 “Upon Arrival”** the customer interacts with the customers already in the queue upon their arrival.
- **Case 2 “After Arrival”** the customer interacts with the customers who arrive while they’re in the system.

These cases are related only through the shared amount of time the customer spends in the system.

Susceptible Model

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Model

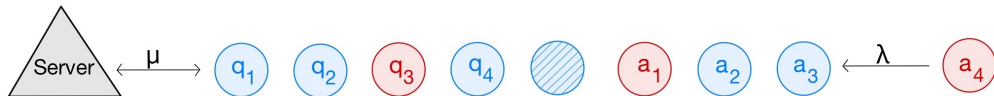
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- Probability p infectious
- Probability $1 - p$ susceptible
- Overlap threshold ℓ
- Susceptible arrival at time t
- Partition into upon and after arrivals

Susceptible Model

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Closest people

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In general, the overlap time between the n^{th} and $n + k^{\text{th}}$ customers is

$$O_{n,n+k} = (S_n + W_n - (A_{n+k} - A_n))^+.$$

Our case methodology makes it useful to hone in on specific overlaps:

Upon Arrivals

- consider all earlier arrivals as customers $1, \dots, k$
- the closest infectious person: the last infectious arrival, $I_{\max} \leq k$

Lemma 1

Supposing queue length k , for all $1 \leq j \leq k$, $\mathbb{P}(I_{\max} = j | Q(t) = k) = p(1 - p)^{k-j}$.

After Arrivals

- consider only the infectious arrivals after us, $1, \dots, k$
- the closest infectious person: the first infectious arrival, A_1

Lemma 2

Supposing queue length k , $\mathbb{P}(A_1 > x - \ell | N_I = k) = \left(\frac{\ell}{x}\right)^k$.

Symmetrical probabilities

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Theorem 3

Let a single susceptible customer join the $M/M/1$ queue in steady state, then the probability the customer gets infected by a customer who is in queue upon their arrival is equal to the probability they get infected by a customer who joins the queue after their arrival,

$$\mathbb{P}(I_{ua}) = \mathbb{P}(I_{aa}) = \frac{p\lambda}{\mu - \lambda(1-p)} e^{-(\mu-\lambda)\ell}. \quad (1)$$

- For a random threshold, one can then integrate over the distribution.
- Note that first term is $\mathbb{P}(S_{\mu-\lambda} > S_{\lambda p}) = \frac{p\lambda}{p\lambda + \mu - \lambda}$.

Probability intersection

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Theorem 4

$$\mathbb{P}(I_{ua} \cap I_{aa}) = \left(\frac{p\lambda}{\mu - \lambda(1-p)} \right)^2 e^{-(\mu-\lambda)\ell}.$$

Intuitive product of three probabilities:

- the probability of an infectious customer overlapping before us,
- the identical probability of an infectious customer overlapping after us, and
- the probability of us being in the system for the infection threshold.

The coupling of the upon- and after-arrival probabilities of infection occurs through the shared requirement that our sojourn time is over the threshold.

Probability you get infected

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Let $I = I_{ua} \cup I_{aa}$ be the event that a susceptible customer gets infected while in the system.

Theorem 5

Suppose a susceptible customer joins the $M/M/1$ queue in steady state, then the probability the customer gets infected is equal to,

$$\mathbb{P}(I) = \frac{p\lambda}{\mu - (1-p)\lambda} \left(2 - \frac{p\lambda}{\mu - (1-p)\lambda} \right) e^{-(\mu-\lambda)\ell}.$$

Invariant to scaling $\rho = \frac{\lambda}{\mu}$, as long as we scale ℓ by $1/n$.

Probability graphs

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Model

Probability you
get infected

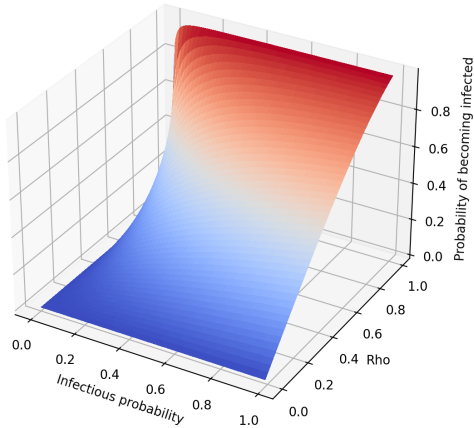
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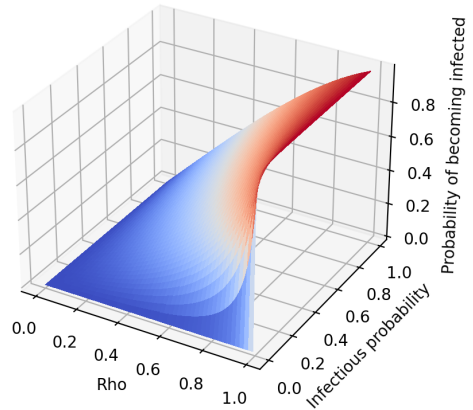
Numerical results

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Infection Probability ($I = 0.5$)



Infection Probability ($I = 0.5$)



Scale invariance

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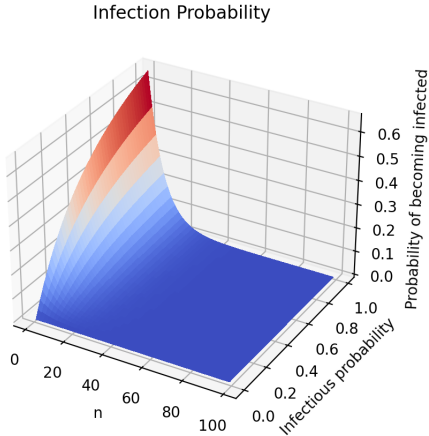


Figure: $\frac{\lambda}{\mu} = 0.5n, \ell = 0.25$

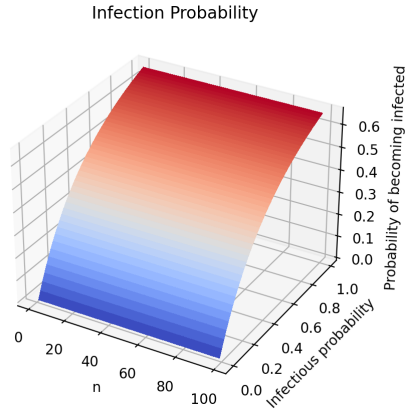


Figure: $\frac{\lambda}{\mu} = 0.5n, \ell = 0.25n$

Reframe problem: dual perspective

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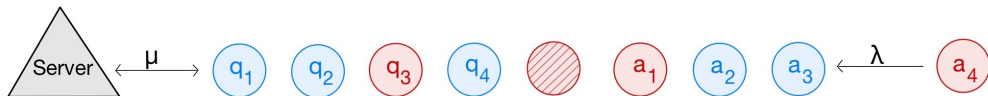
Probability you
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Numerical results

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- Probability p infectious
- Probability $1 - p$ susceptible
- Overlap threshold ℓ
- **Infectious arrival at time t**
- Partition into upon and after arrivals

Reframe problem: dual perspective

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- Probability p infectious
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- **Infectious arrival at time t**
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Reconsider overlaps

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Probability you
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Numerical results

References

Our case methodology again makes it useful to hone in on specific overlaps:

Upon Arrivals

- consider all earlier arrivals as customers $1, \dots, k$
- the furthest customer potentially infectible: the first customer with remaining sojourn time at least ℓ

Lemma 6

Supposing queue length k , for all $0 < j \leq k$, $\mathbb{P}(S_{j>\ell} | Q(t) = r) = \frac{(\mu\ell)^{j-1} e^{-\mu\ell}}{(j-1)!}$.

After Arrivals

- consider only the susceptible arrivals after us
- the furthest susceptible customer: last arrival while we've still got ℓ left in the system

Expectation

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Theorem 7

For a single infectious customer arriving to the system at time t , the expected number of susceptible customers who become infected is

$$\mathbb{E}[\text{Number of New Infections}] = \frac{2\lambda(1-p)}{\mu-\lambda} e^{-(\mu-\lambda)\ell}.$$

- Invariant to scaling $\rho = \frac{\lambda}{\mu}$, as long as we scale ℓ by $1/n$.
- Matches results by Kang et al. [5] when ℓ exponential.

Number of new infections in time t

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References

- $t\lambda(1 - p)$ susceptible arrivals
- $t\lambda p$ infectious arrivals
- Expect $t\lambda(1 - p)\mathbb{P}(I)$ new infections
- Expect infectious arrivals to transmit infection $t\lambda p\mathbb{E}[\# \text{ of new infections}]$ times

Counts multiple transmissions to the same susceptible customer!

- $\frac{\lambda}{\mu} \ll 1$: number of overlaps $\downarrow$, repeat transmissions goes to 0.
- $\frac{\lambda}{\mu} \rightarrow 1$: duplicate transmissions $\uparrow$.

Number of new infections

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Probability you
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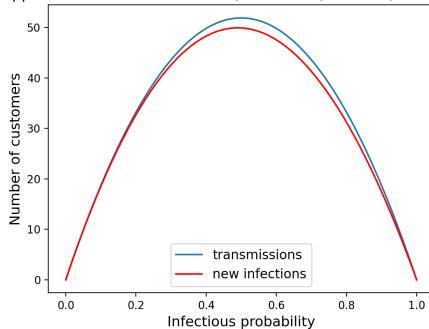
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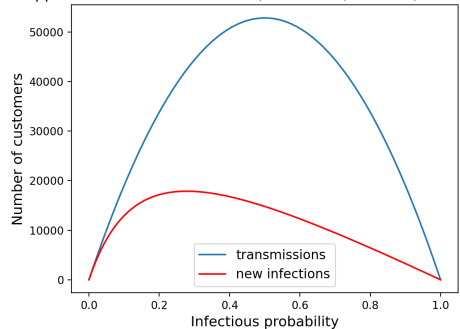
Numerical results

References

Approximate number infected ($\rho = 0.05$, $I = 0.25$, $t = 50000$)



Approximate number infected ($\rho = 0.75$, $I = 0.25$, $t = 50000$)



Second Moment: geometric component

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Lemma 8

Consider a single infectious arrival to the $M/M/1$ queue at time t . The second moment of the number of new infections arising amongst the customers already in the system is equal to the second moment of the number of new infections arising amongst the customers who arrive after this customer,

$$\mathbb{E} [N_{ua}^2] = \mathbb{E} [N_{aa}^2] = \frac{\lambda(1-p)}{\mu-\lambda} \left(1 + \frac{2\lambda(1-p)}{\mu-\lambda} \right) e^{-(\mu-\lambda)\ell}.$$

Note that the moments of N_{aa} and N_{ua} are the product of geometric moments with parameter $\frac{1}{1+\frac{\lambda(1-p)}{\mu-\lambda}}$ and the probability we're in the system for the infection threshold.

Variance

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Theorem 9

If a single infectious arrival to the M/M/1 queue occurs at time t , then the variance of the number of new infections arising while this customer is in the system is,

$$\begin{aligned}\text{Var}[\text{Number of New Infections}] = & \frac{2\lambda(1-p)}{\mu-\lambda} e^{-(\mu-\lambda)\ell} \left(1 + \frac{\lambda(1-p)}{\mu-\lambda} \frac{3\mu^2 - \lambda^2}{(2\mu - \lambda)^2} \right. \\ & \left. + \frac{2\lambda(1-p)}{\mu-\lambda} (1 - e^{-(\mu-\lambda)\ell}) \right).\end{aligned}$$

Numerical results

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Fix $\mu = 1$ and $\ell = 0.5$.

- Average over 5 simulations
- 50,000 customers per simulation
- Note as $\frac{\lambda}{\mu} \rightarrow 1$, the number of customers to reach steady state increases

λ	ρ	prob infected (formula)	prob infected (simulation)	mean infected (formula)	mean infected (simulation)	variance infected (formula)	variance infected (simulation)
.50	.01	0.01535	0.01544	1.54203	1.50240	4.08324	4.79649
.75	.01	0.05066	0.04855	5.24203	5.20114	33.18818	41.38196
.90	.01	0.15060	0.14824	16.95091	17.40381	305.039313	378.01052
.99	.01	0.74375	0.52680	195.04235	107.27622	38237.51635	16959.72723
.50	.02	0.03024	0.02995	1.52645	1.54192	4.01659	5.31996
.75	.02	0.09708	0.09511	5.18908	5.19631	32.57352	39.13737
.90	.02	0.26807	0.24984	16.77969	16.32622	299.07752	295.45574
.99	.02	0.88297	0.75293	193.07222	154.41266	37470.89289	24190.44443

Conclusion

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- Upon and after arrival methodology provides blueprint for analysis
- Methodology allows us to generalize the distribution of the threshold

Where next?

- Infinite servers system does not display same symmetry.
- Case analysis for many server queues becomes challenging.
- Transient behavior of model.
- Allowing a “total contact” threshold measuring a customer’s continuous overlap with *any* infectious customers.
- Allowing a weighted threshold which is reduced if a customer simultaneously overlaps with multiple infectious customers.

References I

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References

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References II

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References

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